Paper Making Machine

Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Hollander Beater, Hydra Pulper, Paper Squeezing Machine, Klin Drying Machine etc; Hydra Pulper (Breaking/recycling paper)

What it is used for

Paper-making equipment processes paper pulp or recycled paper through operations such as pulping, beating, pressing, and drying. Follow the procedure for the specific machine and maintain safe handling of wet pulp, moving parts, and heated surfaces.

Main parts

- Pulping tank or hydra pulper with rotor
- Beater or refining unit, where fitted
- Screen, strainer, or pulp transfer system
- Pressing rollers or squeezing press
- Drying unit, rack, or heated chamber
- Control panel, guards, and emergency stops

Before you start

1. Identify the equipment type and read its operating instructions.
2. Inspect tanks, blades, rollers, belts, guards, hoses, and electrical connections.
3. Check that emergency stops and protective guards are functional.
4. Confirm the pulp mixture, water level, and batch size meet the process requirements.
5. Ensure drains and transfer paths are clear and collection containers are ready.
6. Wear suitable PPE, including gloves and eye protection, and keep the floor dry where possible.

How to use

1. Switch off and isolate power before opening covers or inspecting internal components.
2. Prepare the pulp or recycled paper mixture according to the process instructions.
3. Load the specified amount of material without exceeding the machine capacity.
4. Close covers and secure guards before starting the machine.
5. Select the correct pulping, beating, pressing, or drying mode for the operation.
6. Start the machine and observe its operation for abnormal noise, vibration, leakage, or blockage.
7. Transfer pulp between process stages using the designated equipment and safe handling methods.
8. For sheet forming, distribute pulp evenly according to the machine's forming procedure.
9. Operate the press only with hands and tools clear of rollers and pressure zones.
10. Dry the formed paper using the specified temperature and time, where applicable.
11. Stop the machine before removing material from hazardous areas or clearing a jam.
12. Switch off and isolate power before cleaning or maintenance.

3. Remove pulp residue, clean the equipment, and dispose of waste according to site procedures.
4. Record batch details, operating settings, and any faults.

⚠ Safety

- Only trained personnel should operate pulping, pressing, and drying equipment.
- Never place hands or tools inside a running pulper or between moving rollers.
- Keep guards and interlocks fitted and never bypass emergency stops.
- Isolate and verify zero energy before clearing blockages or servicing machinery.
- Use caution around hot drying surfaces, steam, and heated rollers.
- Handle wet floors and pulp spills promptly to prevent slips; follow chemical safety instructions for additives.

Care & storage

- Clean pulp tanks, screens, transfer lines, and rollers after use.
- Inspect blades, seals, hoses, belts, and guards regularly.
- Remove fibre buildup before it hardens or obstructs the machine.
- Check electrical systems and emergency stops according to the maintenance schedule.
- Arrange servicing for leaks, abnormal vibration, or damaged parts.

Common mistakes to avoid

- Overloading the pulper or using the wrong pulp consistency.
- Clearing blockages while machinery is energised.
- Placing hands near rollers during pressing or sheet removal.
- Using incorrect drying settings or ignoring wet-floor hazards.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.